CHAPTER 07

GRAPHS

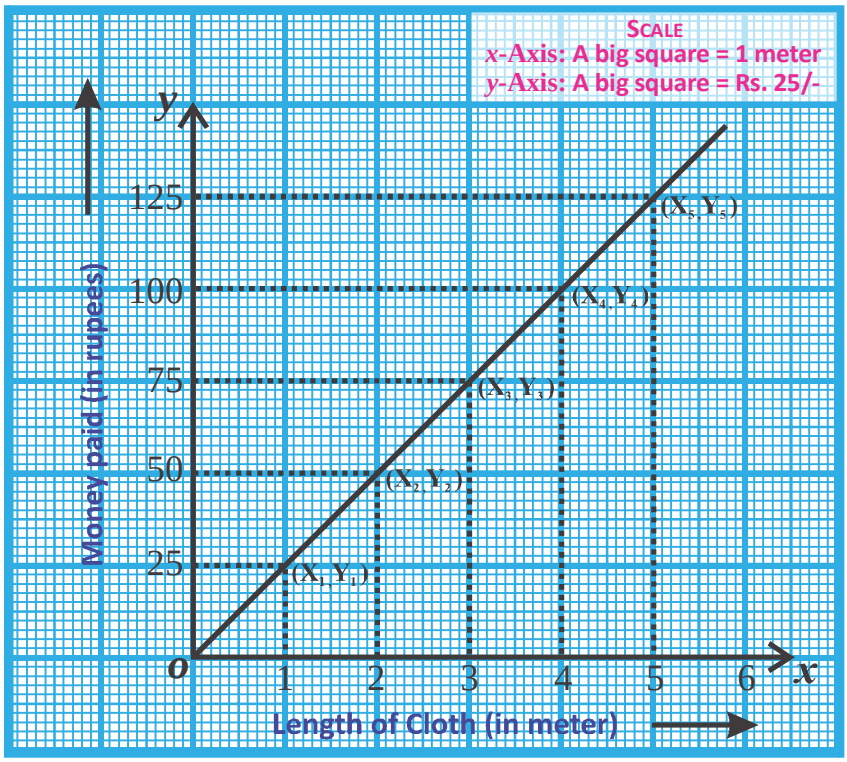
Introduction

In mathematics we use different types of representations, including visuals, to better understand and analyze information. Graphs are one such visual representation and graphs help us define relationship between two quantities. If we have two mutually dependent quantities then we can also use graphs to tell how change in one quantity will lead to change in the second quantity. In addition, drawing graphs can also help in obtaining new information. In this chapter, we will look at different applications of graphs.

Relationship between two quantities

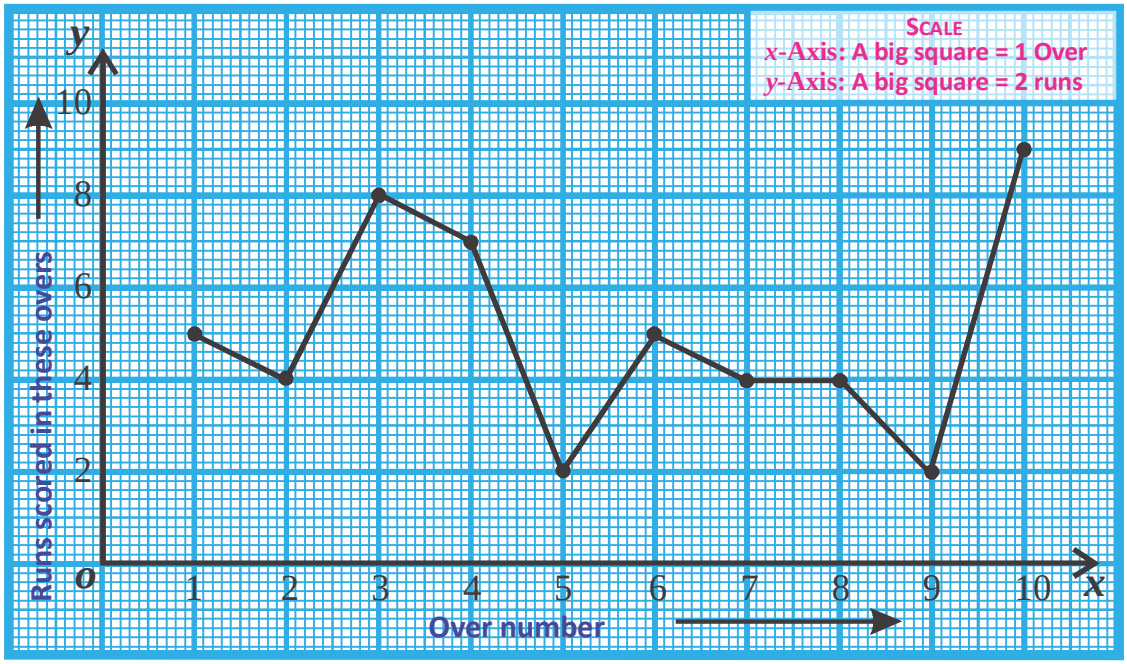
Some people buy 1 meter, 2 meter, 3 meter, 4 meter and 5 meter cloth, at Rs.25 per meter, paying Rs.25, Rs.50, Rs.75, Rs.100 and Rs.125 respectively. The relation between the money spent by them and the length of cloth bought is shown in the graph. From the graph we are able to see how the money spent changes with increase in length of cloth purchased.

Let us see another example.



Graph - 01

The runs scored by a cricket team in the initial ten overs are as follows – 5,4,8,7, 2, 5, 4, 4, 2 and 9. If we take the over number and the runs scored in that over to draw a graph, then we will get the curve shown in figure-2.



Graph - 02

Think and Discuss



1. In each of the situations described above, is there any relation between the quantities taken to draw graphs?
2. One type of quantity was taken on the X-axis and the other on Y-axis. Is there any basis of choosing a particular axis for a particular quantity?
3. One graph is in the form of a straight line and the other is a zig-zag curve; what could be the reason for this difference?

Learning to draw graphs

We require two types of data to draw graphs. One type is depicted on the X-axis and the other on Y-axis. Can we take any axis to show these quantities? Or is there a basis for deciding which quantity should be shown on the X-axis and which on Y-axis?

If we look at the graph-1 we find that if we buy more cloth then we have to pay more money and if we buy less cloth then we spend less money. This is one example where one quantity directly affects the other quantity. The amount of money paid or

spent directly depends on the length of cloth purchased. So we can say that the length of cloth is an independent variable whereas the money spent is dependent variable. Usually, we take the independent variables (data points) on the X-axis and the dependent variables on the Y-axis.

Once we have determined which quantity to keep on the X-axis and which quantity should be taken on the Y-axis then we decide the scale.

Scale – To depict the chosen quantities on the X-axis and Y-axis, we need to choose a suitable scale according to the values of chosen quantitates. Let us understand what this means using graph-1 as example. The payment made for 5 meter cloth is Rs.125. If we decide that in our scale Rs.1 = the length of one square, then our axis would be 125 squares long which is not possible on a sheet of paper. On the other hand, if we take Rs.50 = length of one square, then our axis would become very condensed. So we have to choose a scale that clearly shows the relations. Here, we have taken length of one square = Rs.25 and then drawn a 6 units long axis. While drawing graphs, the following points should be kept in mind for choosing a scale:

- The difference between the maximum and minimum values of each quantity
- Utilizing as much area as possible of the graph paper

Each point has to be shown on the graph. For a given value on the x-axis, we mark a point at a distance equal to the corresponding y value. Both the values determine the position of the point. We get the graph by joining all points.

What information do we get from graphs?

You must have seen different types of graphs in newspapers, magazines and on television. These graphs are visual representations of numerical data. They give us several pieces of information at a glance. Let us look at the two graphs discussed so far. When we plot the graph by taking the length of cloth and its cost on the two axis then the straight line obtained shows that there is a fixed ratio between the cost and length.

$$\left(\frac{100}{4}, \frac{75}{3}, \frac{25}{1} \cdots \text{etc.}\right)$$

If we want to know how much cloth can be purchased for a given amount of money or what will be the cost of given length of cloth, we can find the answers easily from the graph.

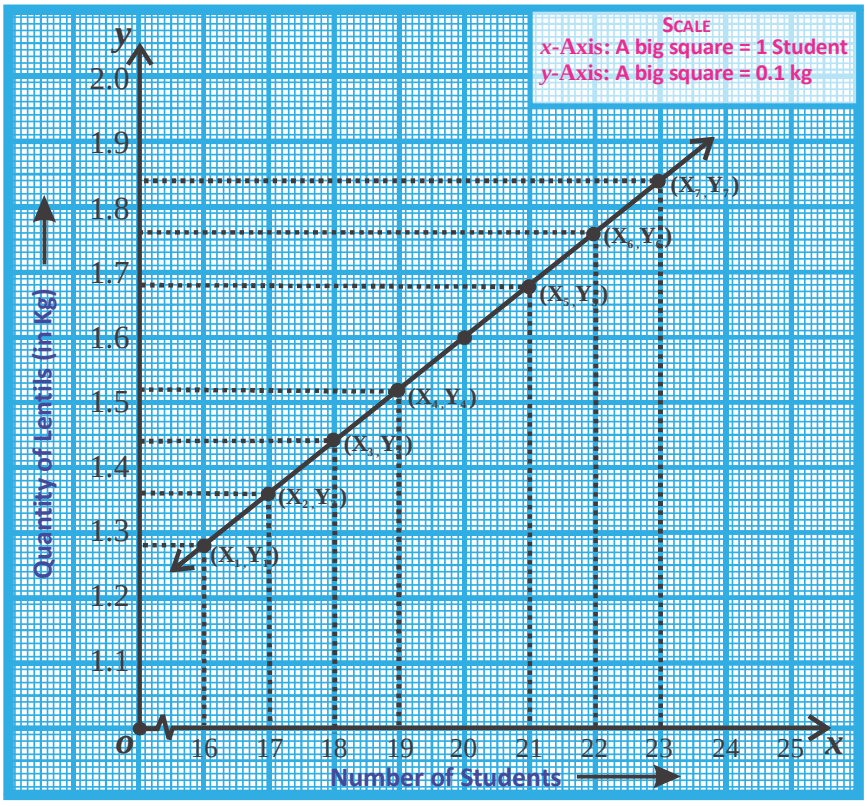
The second zig-zag graph tells us that the number of runs scored in any over is not certain and we can't predict. But we can easily tell the overs in which maximum and minimum runs were scored by looking at the graph. The average runs scored can help us predict how many runs might be scored at the end of 20 or 50 overs. Of course, our prediction may be wrong because the entire team could get out or the rate of scoring could accelerate in the final overs.

Some more graphs

Graph 3:- The following table shows the number of students in the hostel of an upper primary school in Jamraon and the quantity of lentils cooked for them in any given week.

Number of the students	16	19	22	23	21	18	17
Quantity of lentils (in kg)	1.280	1.520	1.760	1.840	1.680	1.440	1.360

Number of students and quantity of lentils are the data points.



Graph - 03

Do you see any relation between the number of students and the quantity of lentils cooked for them? Let us understand by making a graph.

We can see that the quantity of lentils increases and decreases with increase or decrease in number of students. Is there a fixed rate for this change?

Discuss among yourselves. Notice that this graph does not start from 0,0. Often, we find situations where data points do not start from zero or near it. In such a situation we use ⌇ on the axis to denote the empty region of the graph. For example, in graph-3 the readings on x-coordinate start from 16 and there are no data points between zero and sixteen so this portion is shown by a ⌇.



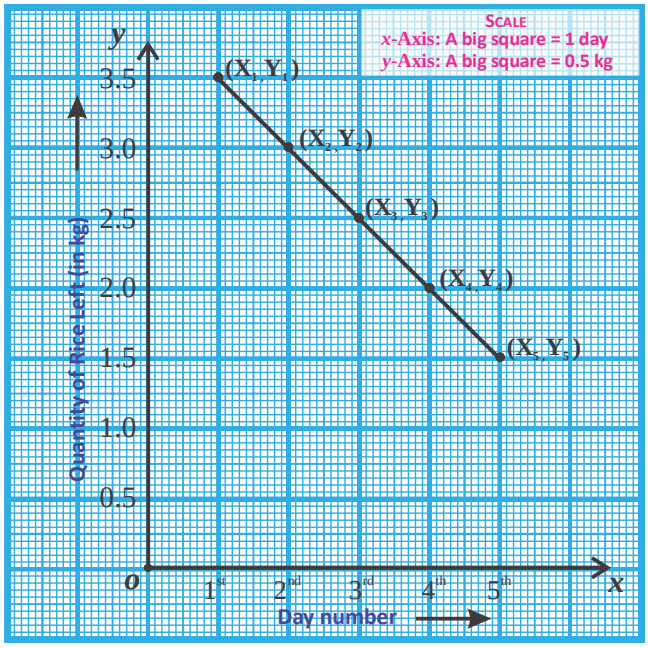
Think and Discuss

If we plot a day-wise graph between number of students and quantity of lentils cooked, what will be the nature of the graph?

Graph-4. Phoolmanti purchased 4 kg rice for her home. 500 g rice is cooked each day in her house. Can we plot the graph for quantity of rice left each day?

Solution : Here the two datas are days and quantity of rice remaining on that day. The first data point is (1,3.5) and the fifth data point is (5, 1.5).

You can see that with increase in number of days, the quantity of rice remaining is decreasing. By looking at the graph can you tell when the rice would finish?



Graph - 04

Try this

1. Collect similar data from around you and plot graphs for this data.
2. Although both graphs 3 and 4 are straight lines yet they are different from each other. What are the differences in the two graphs?
3. Make (x,y) table for graphs 3 and 4.



Graph-5. The table below shows the lengths of one side of some squares and their respective perimeters.

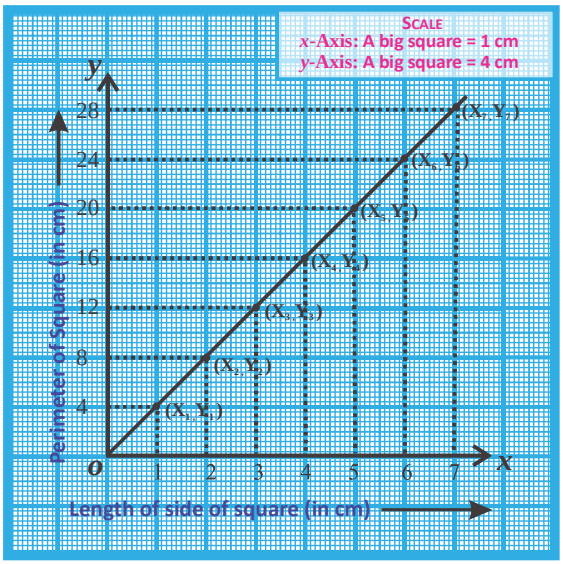
Length of side of square (in cm)	1	2	3	4	5	6	7
Perimeter of square (in cm)	4	8	12	16	20	24	28

Draw a graph between the figures given in the table and answer the following questions –

1. Which data set did you choose for the *x*-axis?
2. Which data set did you choose for the *y*-axis?

Solution : In the graph we can see that increasing the length of any one side of a square leads to increase in perimeter. This means that in the given data-sets, the length of side of a square is an independent variable and perimeter is dependent variable. Therefore, we will take length of side of square on the *x*-axis and perimeter on the *y*-axis.

Graph-6. The lengths of sides of different squares and corresponding areas are shown in the table given below. Use them to draw a graph.



Graph - 05

Length of side of square (in cm)	0	1	2	3	4	5	6	7
Area of square (in cm square)	0	1	4	9	16	25	36	49

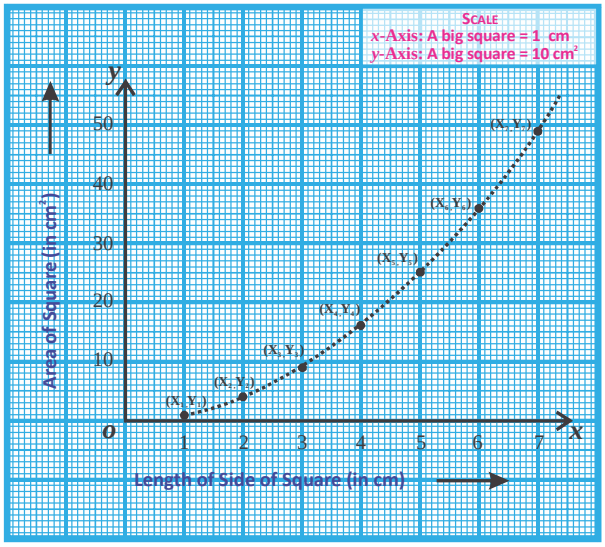
area we find that area increases with increase in length of side but we do not get a straight line. Instead we get an upward moving curve.

Think and Discuss



What differences can you see between graphs 5 and 6? Discuss.

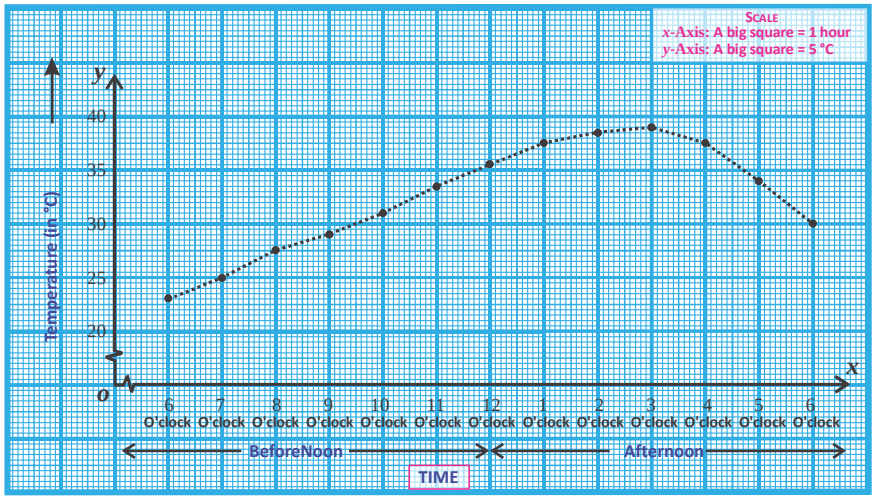
Graph-7. The temperatures between 6 *am* and 6 *pm*, on any given day in the month of March are shown in the table given below.



Graph - 06

Draw a graph based on these figures.

Time	Morning							Afternoon					
	6	7	8	9	10	11	12	1	2	3	4	5	6
Temp. (in°C)	23	25	27.5	29	31	33.5	35.5	37.5	38.5	39	37.5	34	30



Graph - 07

Can you think why this is so?

Discuss with your friends.

Write five conclusions based on this graph.

Graph-8. Find the simple interest on Rs.100 for 1,2,3 and 4 years at 10 percent annual interest rate. Draw the graph between time and

simple interest and see how the interest changes with time.

Also answer the following questions:

1. Which data set did you choose for the x-axis?
2. Which data set did you choose for the y-axis?
3. What scale did you choose for the x- and y-axis?

Solution : We have been given:-

Principal = Rs.100 and Interest = 10%

Keeping the principal and rate of interest constant and changing time period to 1,2,3 and 4 years, we can obtain the simple interest as shown in the table below:

Time (in years)	0	1	2	3	4
Simple Interest (in rupees)	0	10	20	30	40

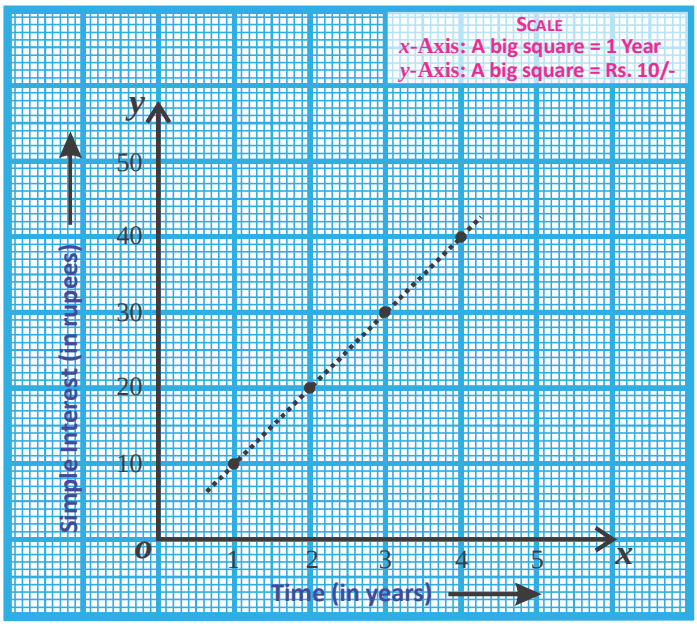
$$\text{Simple Interest} = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$$

From the graph, we can see that when rate and principal are constant then the simple interest changes at a constant rate with change in time period.

We have taken time (independent variable) on the *x*-axis and simple interest (dependent variable) on the *y*-axis.

Scale- On *x*-axis, 1 unit = 1 year

On *y*-axis, 1 unit = Rs.10



Graph - 08

Try This



1. Some persons were loaned Rs.100, Rs.200, Rs.300 and Rs.400 at 10 percent annual simple interest. Draw the graph for the interest obtained from each of them.
2. Make a table showing the age (in months) and height (in cm) of your classmates. Plot a graph between age and height. From this plot, can you see a definite relation between age and height?

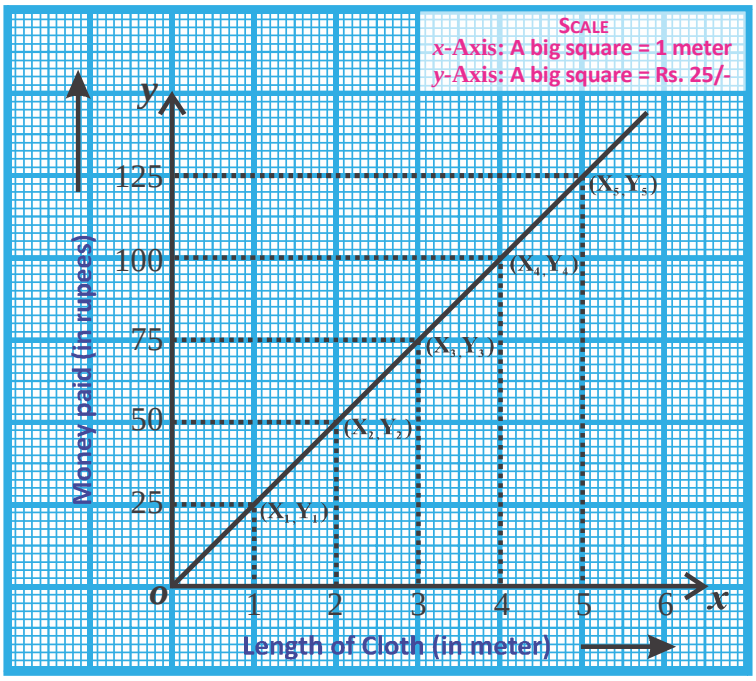
In the graphs discussed so far, we sometimes got a curve and sometimes a straight line. Can you think why?

It is clear that the shape of the plotted graph depends on the relation between the quantities represented on it. This relation also decides the shape of the curve. We will now try to find the relation between different quantities.

In graph-1,

$X_1 = 1,\ X_2 = 2,\ X_3 = 3,\ X_4 = 4,\ X_5 = 5$

$Y_1 = 25,\ Y_2 = 50,\ Y_3 = 75,\ Y_4 = 100,\ Y_5 = 125$



Graph - 01

Here, $\frac{Y_2 - Y_1}{X_2 - X_1} = \frac{50 - 25}{2 - 1} = \frac{25}{1}$

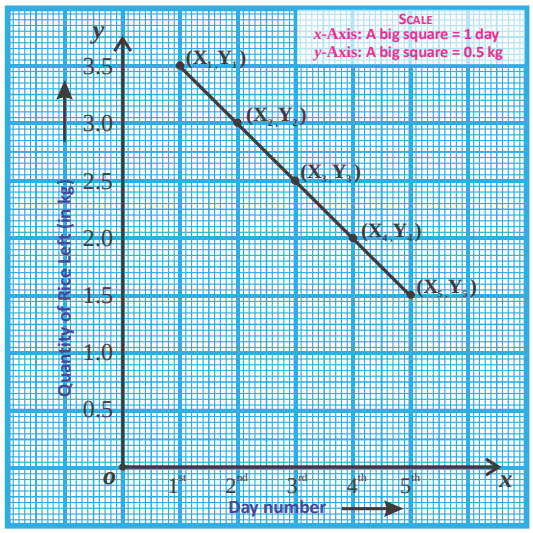
$\frac{Y_3 - Y_2}{X_3 - X_2} = \frac{75 - 50}{3 - 2} = \frac{25}{1}$

$\frac{Y_5 - Y_4}{X_5 - X_4} = \frac{125 - 100}{5 - 4} = \frac{25}{1}$

In graph 4,

$$\frac{Y_2 - Y_1}{X_2 - X_1} = \frac{3 - 3.5}{2 - 1} = -0.5$$

$$\frac{Y_3 - Y_2}{X_3 - X_2} = \frac{2.5 - 3.0}{3 - 2} = -0.5 \quad \text{..... etc.}$$



Graph - 04

In graph 5,

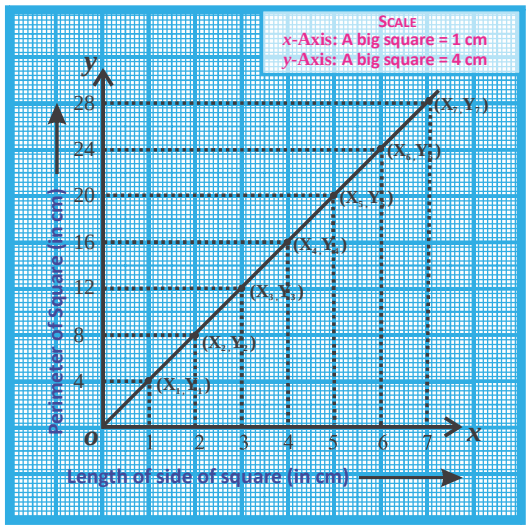
$$\frac{Y_2 - Y_1}{X_2 - X_1} = \frac{8 - 4}{2 - 1} = 4$$

$$\frac{Y_3 - Y_2}{X_3 - X_1} = \frac{12 - 8}{3 - 2} = 4 \quad \text{..... etc.}$$

We can see that each of in graphs 1,4 and 5

$$\frac{Y_2 - Y_1}{X_2 - X_1} = \frac{Y_3 - Y_2}{X_3 - X_2} = \cdots\cdots\cdots\cdots$$ Is constant and the graph is a straight line.

This means that wherever,



Graph - 05

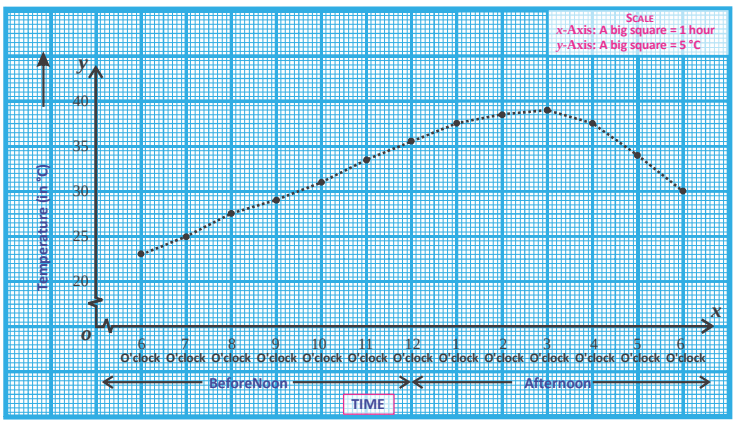
$$\frac{Y_2 - Y_1}{X_2 - X_1} = \frac{Y_3 - Y_2}{X_3 - X_2} = \frac{Y_4 - Y_3}{X_4 - X_3} = \cdots\cdots\cdots = \frac{Y_n - Y_{n-1}}{X_n - X_{n-1}}$$ = is a constant, the graph will be in the form of a straight line. In these types of plots the relation between quantities can be depicted by the linear equation

$ax + by = c$ or $y = mx + c$.

Is there any such relation in the other graphs as well?

In graph-7 we find that-

$$\frac{Y_2 - Y_1}{X_2 - X_1} = \frac{25 - 23}{7 - 6} = \frac{2}{1}$$



Graph - 07

$$\frac{Y_3 - Y_2}{X_3 - X_2} = \frac{27.5 - 25}{8 - 7} = \frac{2.5}{1}$$

$$\frac{Y_4 - Y_3}{X_4 - X_3} = \frac{29 - 27.5}{9 - 8} = \frac{1.5}{1}$$

--- etc.

Clearly,

$$\frac{Y_2 - Y_1}{X_2 - X_1} \neq \frac{Y_3 - Y_2}{X_3 - X_2} \neq \frac{Y_4 - Y_3}{X_4 - X_3} \ldots\ldots\ldots$$

Similar relationship can also be observed in graph-6.

In these two examples,

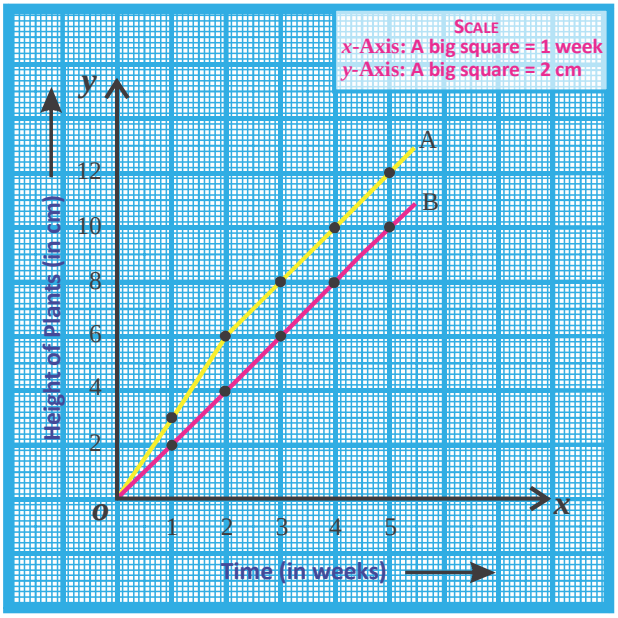
$$\frac{Y_2 - Y_1}{X_2 - X_1}, \frac{Y_3 - Y_2}{X_3 - X_2}, \frac{Y_4 - Y_3}{X_4 - X_3} \ldots\ldots\ldots$$ …are not constant.

Therefore, the plots are not straight lines.



Try this

Note the height of the plants in and around your school for 10 weeks and plot the obtained observations. How did the height of plants change with time?



Graph - 09

Reading graphs describing different situations

Now we will learn how to read, understand and analyze graphs through various examples.

Example:-1. To different types of plants were potted in two pots A and B and their height was noted at the end of the week for 5 weeks. The measurements are shown in the graph. Use the graph to answer the following questions:

(i) What was the height of each plant at the end of five weeks?

(ii) In which did the plant in pot A grow most and what was the increase in its height in this week?

(iii) What was the height of the plant in pot B at the end of week four?

Solutioin : From the graph we can see that,

(i) At the end of five weeks, the height of plant in pot A is 12 cm and that in pot B is 10 cm.

(ii) The height of the plant in pot A increased by 3 cm at the end of two weeks. This was more than the increase in any subsequent week.

(iii) The height of the plant in pot B at the end of week 4 was 8 cm.

Example-2. The following graph shows the runs scored by teams A and B in a 20-20 match. Answer the following questions with the help of the given graph:-

(i) How many runs were scored by team A at the end of over 16?

(ii) What are the intervals when the rates of scoring runs were fastest for team A and team B?

(iii) What are the intervals when the rates of scoring runs were slowest for team A and team B?

(iv) What was the difference in scores of team A and team B at the end of over 8?

(v) By looking at the graph can you tell which team won?

Solution :

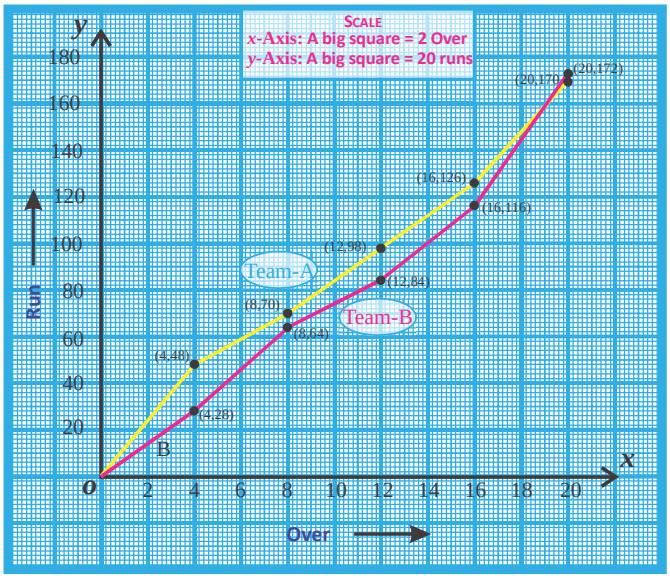
(i) Team A had scored 126 runs by the end of over 16.

(ii) The rates of scoring runs were fastest for team A and team B between overs 16 and 20.

(iii) Team A scored slowest between overs 4 and 8; team B scored slowest between overs 8 and 12.

(iv) The difference in scores of team A and team B at the end of over 8 was 6 runs.

(v) The graph clearly shows that team B won.



Graph - 10



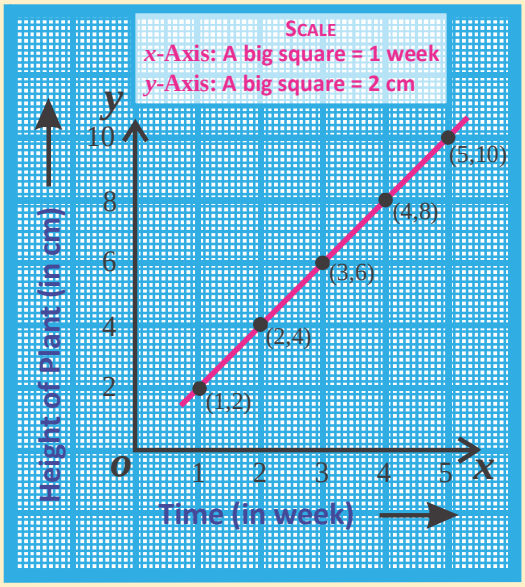
Try this

Show the runs scored by you and your friends in a cricket match on a graph.

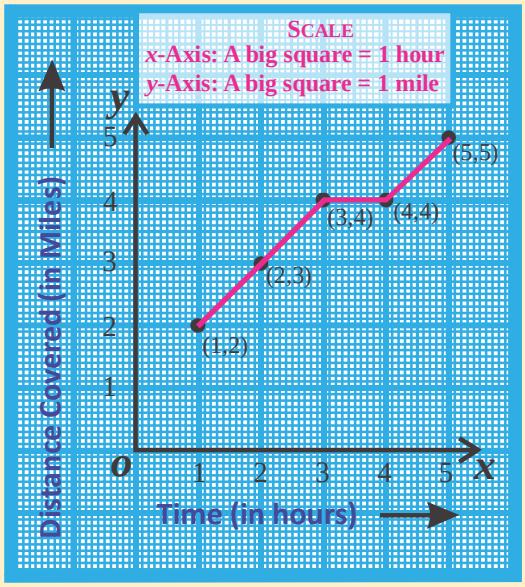
Exercise-1



1. Look at graphs (A) and (B) given below. Graph (A) is a straight line and graph (B) is a curve.



Graph - A



Graph - B

(i) What conclusions can you draw from graph (A)?

(ii) How are these conclusions different from those drawn from graph (B)?

2. A man filled 5 liters of petrol in the fuel tank of his vehicle. The table below shows the days and the petrol remaining on that day-

Day	1	2	3	4	5
Vlume of petrol remaining (*l*)	4	3	2	1	0

Draw a graph between the petrol remaining and number of days.

3. The table below shows simple interests on a deposit of Rs.300 at the interest rate of 5 percent per annum.

Time (in years)	0	1	2	3	4	5
Simple Interest (in rupees)	0	15	30	45	60	75

Draw a graph between time and simple interest.

4. Find the value of x^2 for different values of x and draw a graph between x and x^2. The values of x are integer numbers between -4 and +4.

5. The table below shows the quantities of onions consumed (in kg) over 5 weeks by a family.

Week	1	2	3	4	5
Quantity of onion (in kg)	1	2	3	4	5

Draw a graph between week and quantity of onion consumed.

6. According to an article published in a science magazine the speed of ants in a particular place is effected by the temperature of that place. If equation $s = \frac{t-20}{5}$ shows the relation between the temperature of a place and the speed of ants where temperature, t is in °C and speed, s is in cm per second, then draw the graph between speed of ants and temperature using t = 25°, 30, 35°, 40-

   a. Which quantity was taken on the X-axis?

   b. Which quantity was taken on the Y-axis?

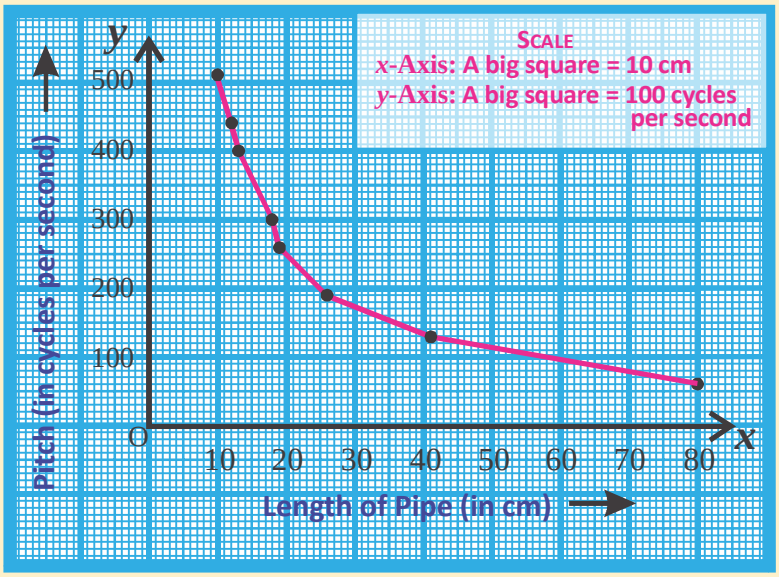
   c. What was the scale used to depict the quantities on x and Y-axis?

   d. What is the temperature of the place where the speed of ants is 2.5 cm per second?

   e. What will be the change in speed of ants if the temperature increases to 40°C from 30°C?

7. Anita made some musical instruments using pipes of different lengths. The mathematical relation between length of pipe (in *cm*) and the pitch (cycles per second) of music is shown in the table below:

Pitch (cycles per second)	64	128	192	261	300	395	438	512
Length of pipe (in cm)	80	41	26	19	18	13	12	10



Look at the graph and answer the following questions:

(i) What should be the length of pipe to obtain a pitch of 160 cycles per second?

(ii) What will be the pitch when the length of pipe is 60 cm?

8. Two tables, A and B are given below. Draw graphs for quantities shown in the tables and find out whether $\frac{Y_2 - Y_1}{X_2 - X_1}$ is constant?

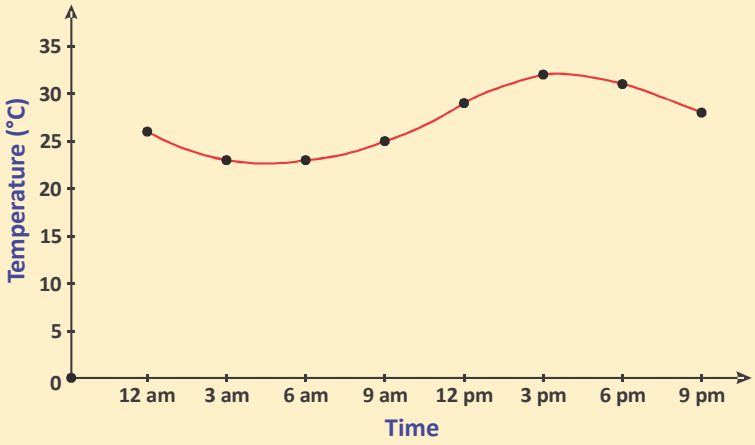
Table - A

	X_1	X_2	X_3	X_4	X_5
Radius of a circle, r (in cm)	2	4	6	8	10
Perimeter of a circle, $2\pi r$ (in cm)	4π	8π	12π	16π	20π
	Y_1	Y_2	Y_3	Y_4	Y_5

Table - B

	X_1	X_2	X_3	X_4	X_5
Radius of a circle, r (in cm)	1	2	3	4	5
Area of a circle, $A = \pi r^2$ (in cm^2)	π	4π	9π	16π	25π
	Y_1	Y_2	Y_3	Y_4	Y_5

9. The temperature data for a particular day in a city are shown in the graph. Answer the following questions based on the graph:



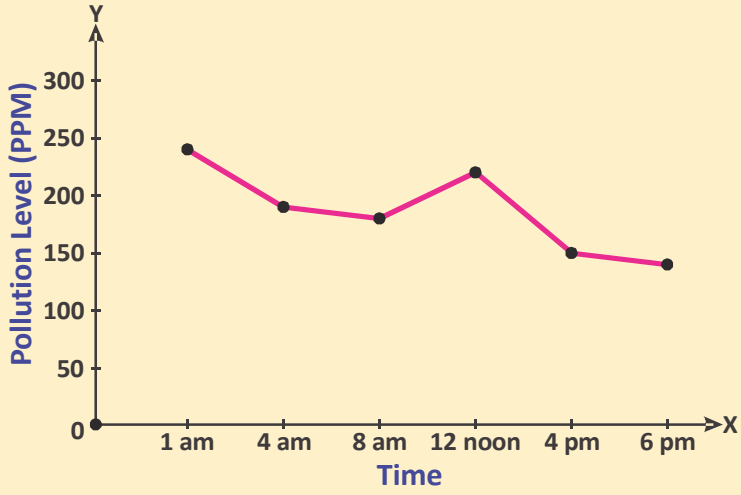
(i) What was the city temperature at 6 am?

(ii) What was the city temperature at 3 pm in the afternoon?

(iii) At what time was the temperature 30°C?

(iv) What would have been the temperature at 12 midnight?

(v) What was the temperature at 9 pm in the night?



10. The pollutions levels of a city were monitored throughout a day. The pollution levels are shown in a graph; answer the following questions based on the graph:-

(i) What was the pollution level at 8 in the morning?

(ii) At what time was the highest pollution level recorded?

(iii) What was the pollution level in the afternoon, at 4 o'clock?

(iv) What was the change in pollution level from 12o'çlock noon to 6 o'clock in the evening?

(v) By how much did the pollution level decrease from 1 o'clock at night to 4 o'clock in the morning?

What we have learnt



1. We learnt to understand the relation between any two quantities using a graph.
2. How to make a graph for given data, how to select the axis for each quantity and how to decide the scale for the graph.
3. Reading graphs describing different situations.
4. Drawing conclusions from a given graph.

